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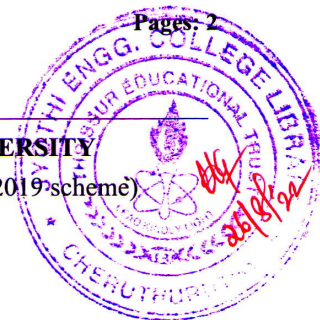
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Reg No.: _____

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fourth Semester B.Tech Degree Examination June 2022 (2019 scheme)



Course Code: EET204

Course Name: ELECTROMAGNETIC THEORY

Max. Marks: 100

Duration: 3 Hours

PART A

(Answer all questions; each question carries 3 marks)

Marks

- | | | |
|----|--|---|
| 1 | Two vectors $\mathbf{A} = 4\mathbf{ax} + 8\mathbf{ay}$ and $\mathbf{B} = 12\mathbf{ay} - 8\mathbf{az}$ are on the same plane. Find the vector normal to the plane. | 3 |
| 2 | Develop the equation for differential volume dv in Spherical coordinate system. Draw necessary figure. | 3 |
| 3 | State Coulomb's law and write the equation in vector form. | 3 |
| 4 | What are equipotential surfaces. Mention two characteristics of equipotential surfaces. | 3 |
| 5 | Derive the expression for $\mathbf{H}$ at a point due an infinitely long conductor carrying a current I . Use Biot Savart's law. | 3 |
| 6 | State Ampere's circuital law for steady magnetic fields. Write the point form of Ampere's circuital law. | 3 |
| 7 | Describe the characteristics of uniform plane waves. | 3 |
| 8 | What is meant by a Poynting vector. Mention its significance. | 3 |
| 9 | What are the primary constants of Transmission line. Draw a neat figure to represent it. | 3 |
| 10 | Define characteristic impedance of a lossless line. | 3 |

PART B

(Answer one full question from each module, each question carries 14 marks)

Module -1

- | | | |
|----|---|---|
| 11 | a) Find the Vector directed from (2, -5, -2) to (14, -5, 4) in Spherical coordinate system. | 7 |
| | b) Define Gradient of a Vector and Mention three characteristics of it. | 7 |
| 12 | a) State Divergence of a vector field and Derive the expression for Divergence of a vector field in integral and point forms. | 7 |
| | b) Derive Coordinate transformation between Cartesian to Cylindrical Coordinates with necessary figure. | 7 |

Module -2

- 13 a) State Gauss's law and obtain Electric field intensity E at any point due to an infinite sheet of charge with the help Gauss's law. Draw necessary figures. 7
- b) Obtain the Potential at a point $(1,2,5)$ due to a dipole with centre at origin and of dipole moment $P = 20\text{Cm}$ acting along the Z axis. Also find the Potential at point $(1,2,0)$. 7
- 14 a) Find the Voltage between two infinitely long conductors parallel to z axis with charges $+4\text{nC}$ and -4nC are passing through points $(0,-4,0)$ and $(0,4,0)$ respectively. Assume free space. 7
- b) Derive the expression for Capacitance per unit length between two infinitely long conductors. 7

Module -3

- 15 a) List all the Maxwell's equation for static Electric and Magnetic fields in Integral and Point forms. Write the significance of each in one sentence. 7
- b) An uniform electric field $E_1 = 5ax - 2ay + 4az$ incidents at the boundary between two dielectrics of permittivity $\epsilon_1 = 2$ and $\epsilon_2 = 4$ respectively. Find the Vector E_2 . Assume the boundary is at $Z=0$ plane. 7
- 16 a) Derive the expression for modified form of Ampere's circuital law. 7
- b) The magnetic flux density at a point which is at a perpendicular distance of 2m from an infinitely long current carrying conductor along Z axis is 0.05 mWb/sq.m . Find the value of dc current passing through the conductor. Also find the value of B at point $(1, 2, 5)$ during the same condition as above. Assume free space. 7

Module -4

- 17 Derive Poynting Theorem. Describe each term in the equation with one sentence. 14
- 18 Calculate the intrinsic impedance η , the propagation constant γ , and the wave velocity u at a frequency $f = 50\text{ Hz}$ for a good conducting medium in which $\sigma = 58\text{ S/m}$ and $\mu_r = 1$. 14

Module -5

- 19 Describe the terms; a) SWR. b) Impedance matching c) Propagation constants. 14
- 20 Derive the wave equations for Voltage and Current in a transmission line with neat figure. 14
